MARLON DE PAZ

AI Infrastructure & Network Architect

CCIE · CKA · NVIDIA Agentic AI Professional · IEEE Co-author

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PROFESSIONAL SUMMARY

AI Infrastructure & Network Architect with 13+ years of global ICT experience, currently architecting the network for sovereign AI cloud infrastructure at Core42 (G42 Group) in Abu Dhabi — Data Center Interconnect across regions, object storage load balancing for AI data pipelines (including 4.8 Tbps GSLB validation), and production agentic AI operations on Kubernetes/OpenShift. Former Nokia specialist in lossless Ethernet fabrics for AI/ML (RoCEv2, ECN, DCQCN, PFC) and Data Center Fabric automation, providing architecture and design consultancy to global enterprise and service provider customers. NVIDIA-Certified Professional: Agentic AI and NVIDIA-Certified Associate: AI Infrastructure & Operations, with NVIDIA training in InfiniBand, RDMA and UFM; CCIE, CKA, Isovalent 20× Lab Champion, and IEEE and Springer co-author. Builds governed agentic AI pipelines (LangGraph proposer–critic, human-in-the-loop) for network operations. Trilingual (English, Spanish, Portuguese). Pursuing CCIE Automation and ISO/IEC 42001 Lead Implementer.

PUBLICATIONS

Co-authored as mentor to the master's students who carried out the research.

Kubernetes-Aware Load Balancing for Leaf-Spine EVPN Data Centers
Journal of Cloud Computing · Springer Nature · 2025 · link.springer.com/article/10.1186/s13677-025-00814-0

Decentralized and Dynamic Zero-Touch Provisioning of Leaf-Spine EVPN Data Centers
IEEE Xplore · ieeexplore.ieee.org/document/11080420

CORE TECHNICAL DOMAINS

Agentic AI & LLM Engineering LangChain · LangGraph / StateGraph · Multi-Agent (Proposer–Critic) · AGNTCY (A2A, SLIM) · MCP · LiteLLM · Claude Code · RAG · Prompt Engineering · Human-in-the-Loop Governance	**AI Engineering Practices** Spec-Driven Development · Harness Engineering · Loop Engineering · Deterministic + Agentic Pipelines · Eval & Guardrails · OpenTelemetry for Agents
Automation & CI/CD Ansible · Terraform · Python · GitLab CI/CD · Test Automation · Event-Driven Automation · NetBox	**Network Architecture & NOS** IP/MPLS · SR-MPLS · EVPN · VXLAN · BGP · Cumulus Linux · SONiC · Mellanox Onyx · SR Linux · gNMI · containerlab · ECN/DCQCN · PFC
AI / HPC GPU Fabric InfiniBand · RDMA · RoCEv2 · NVLink · NVSwitch · GPUDirect · CUDA · DOCA · NVIDIA UFM · OFED	**Cloud & Containers** Kubernetes · Cilium/eBPF · Go (K8s controllers) · OpenStack · Proxmox · Prometheus · Grafana · Loki · Tempo · Zabbix
Security & Compliance NIST CSF · AAA/TACACS+/RADIUS · Red Hat IDM · Secure-by-Design · IAM	**AI Governance (In Progress)** ISO/IEC 42001 · NIST AI RMF · Responsible AI · Risk Assessment · AIMS

PROFESSIONAL EXPERIENCE

Network Architect — Data Center Interconnect (DCI)

Core42 (G42 Group) · Abu Dhabi, UAE · *Dec 2024 – Present*

Architecting the Data Center Interconnect for the sovereign AI cloud of the UAE's national AI platform — one of the largest AI compute environments outside the United States.

- Define and own the end-to-end Data Center Interconnect (DCI) architecture linking the sovereign AI cloud's data centers across regions — an SR-MPLS transport core with SR Traffic Engineering for path control, TI-LFA and BGP-PIC for fast failure recovery, and multi-site EVPN interconnect per IETF RFC 9014 for secure, scalable Layer 2/Layer 3 extension between sites

- Architected the production-grade agentic reporting solution for IP Core and DCI capacity planning — a LangChain / StateGraph pipeline pairing a proposer and critic agent as governance gates over a deterministic data core, with the LLM (GPT-5 via the Compass gateway) generating validated analysis and narrative under human-in-the-loop review. Produces the weekly IP Core capacity consumption report and the executive capacity-planning report for the entire DCI across regions; I deployed it myself on an OpenShift cluster, where it runs autonomously on Kubernetes CronJobs, and hardened both the agent and its CI/CD pipeline against vulnerabilities
- Implemented an agentic operations framework that correlates incidents with change requests using MCP, Zabbix and Jira, helping operators link incidents to the changes behind them and improving cloud operations
- Built a framework to parse all cloud circuits into NetBox using Ansible and REST APIs, ensuring flawless source-of-truth integration across the estate
- Delivered the end-to-end Zabbix observability stack for the IP Core and DCI across Cisco NCS platform variants — from collector deployment and custom template development to low-level discovery, item and trigger prototypes, alarm thresholds, graphs, and NOC dashboards — giving full real-time visibility into core and interconnect health
- Applied event-driven automation to automatically mitigate OpenStack failures in the DCN, reducing mean-time-to-recovery through closed-loop remediation
- Defined the observability stack for the Object Storage Load Balancers (OSS) on Grafana and Alertmanager, designing alert rules and routing driven by labels and annotations for precise NOC signal routing
- Created an orchestration framework to operate the load balancer fleet, enabling activities such as hardening and halting real servers seamlessly and without service impact, as well as TLS certificate service management and rotation
- Defined the Layer 7 ACL model for the load balancers — inspecting HTTP headers and applying a policy action based on the match
- Technical owner for Identity and Access Management (IAM) across multi-vendor environments; designed and govern end-to-end AAA frameworks using Red Hat IDM, TACACS+, and RADIUS with Vendor-Specific Attributes
- Led UAT for 4.8 Tbps Global Server Load Balancer (GSLB) supporting large-scale Object Storage for AI data pipelines
- Applied NIST Cybersecurity Framework secure-by-design principles across all network architecture, deployment, and operational phases
- Selected as AI Champion — responsible for designing and implementing AI-native operations across the organization
- Finalist, G42 Agentathon 2026 — AgentES (Autonomous SDLC): a six-agent software-development-lifecycle simulator built on the AGNTCY platform

Network Architect — IP/MPLS & Data Center Fabric

Nokia · Amadora, Portugal · *Apr 2019 – Nov 2024*

Customer-facing architecture and design consultation across Data Center Fabric, lossless AI/ML Ethernet fabrics, Network Automation, and IP/MPLS Core for global enterprise and service provider clients.

- Responsible for the Smart Grid account portfolio, driving sales of 6,000+ routers and generating more than $6M in revenue over nearly six years
- One of Nokia's prime specialists for lossless Ethernet AI fabrics, working hands-on with RoCEv2, ECN, DCQCN, and PFC congestion management
- Contributed to Data Center Fabric Blueprint testing new features (EVPN IP Aliasing, Event Handler, NetOps Development Kit) using Test Automation and IaC for lab operations
- Designed and owned GitLab CI/CD pipelines for the Fabric Builder framework and its test-automation suite — defining the automated test phase (linting, unit, integration, and end-to-end fabric validation) to enforce quality gates and continuous delivery of validated network builds
- Pioneered Network Automation and Infrastructure-as-Code practices within Nokia SBU, reducing manual configuration time across migration windows for multi-vendor clients
- Supported global fabric and automation initiatives including CNV (Cloud Network Verification), Secure ZTP, and DTaaS (Digital Twin as a Service)
- Delivered customized IP/MPLS network solutions for Energy, Rail, Oil, and critical infrastructure sectors — Network Core, Mobile Backhauling, EPC Infrastructure, residential services, and business VPNs
- Contributed to advanced DC Fabric and quantum-safe AnySec demonstrations — featured at MPLS SD & AI Net World 2024 — and co-organized the 1st Europe ENT DCF Partners Hackathon
- Built a Tier-3 fabric streaming-telemetry stack (Nokia SR Linux + SR OS), published in Nokia's public srl-labs catalogue — github.com/srl-labs/srl-sros-telemetry-lab
- Mentored master's degree students at IST Lisbon University on advanced networking (SRLinux, FSS, Kubernetes) supporting dissertation development

Network Consulting Engineer

Layer2 Consulting LLC (Cisco Contractor) · Honduras · *Mar 2018 – Apr 2019*

- Supported Cisco Advanced Services Team delivering complex solutions for enterprise clients across LATAM; led highly complex upgrade and migration activities across multiple Cisco platforms

– Built simulated networks in test labs to validate new solutions before production; reduced downtime through proactive infrastructure monitoring

Network & Security Infrastructure Specialist

S&H Support (Citi Bank Contractor) · San Salvador · *Feb 2016 – Feb 2018*

– Designed and implemented network and security solutions across LATAM, Caribbean, and Brazil for Citi Bank covering IP Routing Core/Aggregation/Access, Wireless, VOIP, B2B connectivity, NAT, and Firewall
– Led LATAM/Brazil divestiture infrastructure separation procedure — awarded by senior management for delivering ahead of schedule

Telecommunications Engineer

Smart Collections (Tigo Contractor) · San Salvador · *Mar 2014 – Feb 2016*

– Planned and implemented unified service provider networks across IP-MPLS, CORE, IPRAN, HFC, SONET/SDH, and DWDM in multivendor environments (Alcatel, Juniper, Huawei, Cisco)
– Led hierarchical VPN and IGP restructuring for All-IP MPLS mobile backhaul; implemented L2VPN, L3VPN, VPLS, VLL

QA Technician — Carrier Access Deployments

Tech Americas · San Salvador · *May 2013 – Feb 2014*

– Performed quality assurance on T1, T3 and optical fiber customer deployments for US carriers including AT&T and Verizon, verifying that CPE installed by field technicians met design and service standards

SELECTED OPEN-SOURCE PROJECTS

Personal projects and contributions to Nokia's public srl-labs lab catalogue. Full list at github.com/mairp and mairp.ai.

– **agentic-netops** — Intent-to-fabric agentic NetOps: an AGNTCY and LangGraph multi-agent tier turns plain-language intent into Kubernetes resources; Go controllers reconcile them onto a live SONiC EVPN/VXLAN fabric, and gNMI telemetry closes the loop. Nothing is reported as deployed until it is verified on the device. Nokia SR Linux variant: agentic-netops-srl.
– **specstride** — Spec-driven autonomous coding orchestrator: drives an agent through Spec Kit phases behind an independent LLM proposer–critic gate, diagnoses stuck phases, and tunes its own per-phase settings from run telemetry. Python standard library only; companion project mixture-of-loops generates unattended pipelines.
– **agent-observability-stack** — Self-hosted token and cost telemetry for multi-agent fleets: LiteLLM and OpenTelemetry into Prometheus, Loki, Tempo and Grafana, with Arize Phoenix for span-level LLM traces and budget alerts.
– **srl-sros-telemetry-lab** — Streaming-telemetry lab for a Nokia SR Linux and SR OS Clos fabric (three leaves, two spines, two DC gateways) built with containerlab, gNMIc, Prometheus and Grafana; published in Nokia's public srl-labs catalogue.
– **sros-anysec-macsec-lab** — (contributor) Quantum-safe ANYSec and MACsec demo on Nokia SR OS FP5: VLL, VPLS and VPRN services each mapped to its own SR-ISIS / Flex-Algo slice, with gNMIc, Prometheus and Grafana telemetry, EdgeShark capture and a Python/Flask automation panel.
– **kind-cilium-hubble-cluster** — Observable Kubernetes cluster bootstrapped from scratch with kind, Cilium and Hubble.

CERTIFICATIONS & CREDENTIALS

ACTIVE

◆ **CCIE — Cisco Certified Internetwork Expert** *2019*
◆ **Certified Kubernetes Administrator (CKA)** *2019*
◆ **NVIDIA-Certified Associate: AI Infrastructure & Operations** *2025*
◆ **NVIDIA-Certified Professional: Agentic AI** *2026*
◆ **Isovalent 20× Lab Champion (eBPF / Cilium)**
◆ **Anthropic Academy — Claude with the Anthropic API** *2026*
◆ Cisco Certified DevNet Professional *2022*
◆ CCNP Data Center *2025*
◆ Nokia Data Center Fabric Professional *2021*
◆ Cisco DevNet Specialist — Core *2022*
◆ CCNP Enterprise *2015*
◆ JNCIA-Junos — Juniper Networks *2015*
◆ Cisco Certified DevNet Associate *2020*
◆ Cisco Specialist — Service Provider Core *2018*
◆ Cisco Specialist — Enterprise Core *2015*
◆ Cisco Specialist — Enterprise Adv. Infrastructure Impl. *2013*

IN PROGRESS — 2026/2027

- **CCIE Automation (Lab exam: Nov 2026)**
- **ISO/IEC 42001 Lead Implementer (Expected 2026)**
- **Claude Certified Architect — Foundations (Expected 2027)**

NVIDIA AI / HPC FABRIC TRAINING — INFINIBAND · RDMA · UFM

- AI Infrastructure and Operations Fundamentals
- RDMA over Converged Ethernet (RoCE) from A to Z
- Data Center Management Made Easy with NVIDIA UFM
- InfiniBand Network Administration
- InfiniBand Professional

EDUCATION

Bachelor of Engineering — Electrical Engineering
Central American University 'José Simeón Cañas' · San Salvador · 2009

AWARDS & COMMUNITY

- Finalist — 1st G42 Agentathon 2026 (AgentES, Autonomous SDLC)
- AI Champion — Core42 (AI-native operations across the organization)
- Co-author — Springer Nature, Journal of Cloud Computing (2025) & IEEE Xplore (2026)
- Isovalent 20× Lab Champion — eBPF/Cilium advanced networking community recognition
- 1st Place — IEEE Championship: Wireless Control of Mini Cars (2007)
- Contributor — quantum-safe AnySec demonstration featured at MPLS SD & AI Net World 2024
- Built Tier-3 fabric streaming-telemetry lab (SR Linux + SR OS), published in Nokia's srl-labs catalogue — github.com/srl-labs/srl-sros-telemetry-lab
- SRExperts Contributor — Nokia advanced DC Fabric demonstrations
- Co-organizer — 1st Europe ENT DCF Partners Hackathon
- Academic Mentor — IST Lisbon master's students (SRLinux, FSS, Kubernetes)

LANGUAGES

English (Professional) **Spanish** (Native) **Portuguese** (Professional)

Full portfolio: **mairp.ai — AIOps Architect** · *Abu Dhabi, UAE* · *References available upon request*